VERDICT · DFIR CASE FILE

VERDICT — Forensic Investigation Report

DFIR at machine speed · signed, replayable chain of custody

Case ID: `bb2953cd-bdd9-4251-a493-70c97663f892` **Run ID:** `auto-1781370852` **Started:** 2026-06-13T17:14:12Z **Finalized:** 2026-06-13T17:14:22Z
Evidence: `/home/assessor/Desktop/PUG-Projects/sans-hackathon/evidence/SCHARDT.dd` **Verdict:** SUSPICIOUS

Cryptographic attestation: Merkle root `1a2b2f39dd4d8a03e00aae54cd7ace5441eb510f9219dbed593d9e2ded98072a` Audit log final hash `42edd1aa1402b0d1002c9349010f99ad614d54bb9ef204530b65198ee6ef21a9` Ed25519 signature SHA-256 `1f438b48930bf5316c8ff2aa79dfb160efba74b5727c32f962785226bef96d18` Cert fingerprint `b98df1a9d09da3741e295d7da21b9b675287adfb36b10ca17c280e2a1fee0f54`

BOTTOM LINE UP FRONT

SUSPICIOUS CERTAINTY · HIGH

Confirmed: cain.exe executed.

SAM · T1136.001

8 Confirmed 19 Hypothesis

Verdict: SUSPICIOUS. Confirmed: cain.exe executed.

The supplied evidence shows cain.exe executed at 2004-08-27T15:33:03Z.

Scope: findings span 20 hosts — ETHEREAL.EXE-1C148EEF.pf (confirmed); CAIN.EXE-23D61279.pf (confirmed); ETHEREAL-SETUP-0.10.6.EXE-1D932600.pf (confirmed); NETSTUMBLER.EXE-0BFEE568.pf (confirmed); LOOKATLAN.EXE-1F991DD9.pf (confirmed); MIRC.EXE-0661EC22.pf (confirmed); MIRC612.EXE-02791C37.pf (confirmed); CAIN25B45.EXE-056F3A6E.pf (confirmed); NTUSER.DAT (hypothesis); target.lnk (hypothesis); NETSTUMBLERINSTALLER_0_4_0.EX-0BD9920C.pf (hypothesis); SAM (hypothesis); $MFT (hypothesis); Anonyymizer.lnk (hypothesis); channels (2).lnk (hypothesis); channels.lnk (hypothesis); GhostWare.lnk (hypothesis); keys.lnk (hypothesis); Temp on m1200 (4.12.220.254).lnk (hypothesis); INFO2 (hypothesis). Each is assessed separately below; the evidence does not establish them as one incident.

Assessment: The cited tool output meets a defined detection rule for this technique. Treat single-source signals as leads until corroborated across artifact classes.

Certainty: High — the cited tool output is reproducible (the verifier re-ran it and the SHA-256 matched). The confidence is in the artifact, not in intent or actor.

Key findings:

- Suspicious activity (CONFIRMED, T1588.002, cited by tc-010).
- Suspicious activity (CONFIRMED, T1588.002, cited by tc-011).
- Suspicious activity (CONFIRMED, T1040, cited by tc-020).
- Suspicious activity (CONFIRMED, T1040, cited by tc-021).
- Suspicious activity (CONFIRMED, T1046, cited by tc-038).
- Suspicious activity (CONFIRMED, T1071.001, cited by tc-040).
- Suspicious activity (CONFIRMED, T1071.001, cited by tc-041).
- Suspicious activity (CONFIRMED, T1046, cited by tc-047).
- Findings: 27 total — 8 confirmed, 0 inferred, 19 hypothesis.
- Most important next step: Collect Security, Sysmon, and PowerShell Operational EVTX and rerun EVTX/Hayabusa analysis.

HOST ANALYSIS

Findings span more than one host; each is assessed on its own evidence below. The evidence does not establish them as a single incident.

ETHEREAL.EXE-1C148EEF.PF

1 finding(s) — 1 confirmed, 0 inferred, 0 hypothesis · 1 events · 2004-08-27T15:34:54Z · source: ETHEREAL.EXE-1C148EEF.pf

Other: T1040 [CONFIRMED] tc-021

The cited tool output meets a defined detection rule for this technique. Treat single-source signals as leads until corroborated across artifact classes.

TIME (UTC)	EVENT	ACCOUNT	TOOL CALL
2004-08-27T15:34:54Z	prefetch run: ETHEREAL.EXE	—	tc-021
2004-08-27T15:46:13Z	UserAssist records execution of cain.exe	—	tc-214
2004-08-27T15:46:13Z	UserAssist records execution of cain25b45.exe	—	tc-214
2004-08-27T15:46:13Z	UserAssist records execution of ethereal-setup-0.10.6.exe	—	tc-214
2004-08-27T15:46:13Z	UserAssist records execution of ethereal.exe	—	tc-214
2004-08-27T15:46:13Z	UserAssist records execution of lookatlan.exe	—	tc-214
2004-08-27T15:46:13Z	UserAssist records execution of mirc.exe	—	tc-214
2004-08-27T15:46:13Z	UserAssist records execution of mirc612.exe	—	tc-214

CAIN.EXE-23D61279.PF

1 finding(s) — 1 confirmed, 0 inferred, 0 hypothesis · 9 events · 2004-08-27T15:33:03Z → 2004-08-27T15:46:13Z · source: CAIN.EXE-23D61279.pf

Other: T1588.002 [CONFIRMED] tc-010

The cited tool output meets a defined detection rule for this technique. Treat single-source signals as leads until corroborated across artifact classes.

TIME (UTC)	EVENT	ACCOUNT	TOOL CALL
2004-08-27T15:33:03Z	prefetch run: CAIN.EXE	—	tc-010
2004-08-27T15:46:13Z	UserAssist records execution of cain.exe	—	tc-214
2004-08-27T15:46:13Z	UserAssist records execution of cain25b45.exe	—	tc-214
2004-08-27T15:46:13Z	UserAssist records execution of ethereal-setup-0.10.6.exe	—	tc-214
2004-08-27T15:46:13Z	UserAssist records execution of ethereal.exe	—	tc-214
2004-08-27T15:46:13Z	UserAssist records execution of lookatlan.exe	—	tc-214
2004-08-27T15:46:13Z	UserAssist records execution of mirc.exe	—	tc-214
2004-08-27T15:46:13Z	UserAssist records execution of mirc612.exe	—	tc-214

ETHEREAL-SETUP-0.10.6.EXE-1D932600.PF

1 finding(s) — 1 confirmed, 0 inferred, 0 hypothesis · 1 events · 2004-08-27T15:28:36Z · source: ETHEREAL-SETUP-0.10.6.EXE-1D932600.pf

Other: T1040 [CONFIRMED] tc-020

The cited tool output meets a defined detection rule for this technique. Treat single-source signals as leads until corroborated across artifact classes.

TIME (UTC)	EVENT	ACCOUNT	TOOL CALL
2004-08-27T15:28:36Z	prefetch run: ETHEREAL-SETUP-0.10.6.EXE	—	tc-020
2004-08-27T15:46:13Z	UserAssist records execution of cain.exe	—	tc-214
2004-08-27T15:46:13Z	UserAssist records execution of cain25b45.exe	—	tc-214
2004-08-27T15:46:13Z	UserAssist records execution of ethereal-setup-0.10.6.exe	—	tc-214

TIME (UTC)	EVENT	ACCOUNT	TOOL CALL
2004-08-27T15:46:13Z	UserAssist records execution of ethereal.exe	—	tc-214
2004-08-27T15:46:13Z	UserAssist records execution of lookatlan.exe	—	tc-214
2004-08-27T15:46:13Z	UserAssist records execution of mirc.exe	—	tc-214
2004-08-27T15:46:13Z	UserAssist records execution of mirc612.exe	—	tc-214

NETSTUMBLER.EXE-0BFEE568.PF

1 finding(s) — 1 confirmed, 0 inferred, 0 hypothesis · 1 events · 2004-08-27T15:12:35Z · source: NETSTUMBLER.EXE-0BFEE568.pf

Other: T1046 [CONFIRMED] tc-047

The cited tool output meets a defined detection rule for this technique. Treat single-source signals as leads until corroborated across artifact classes.

TIME (UTC)	EVENT	ACCOUNT	TOOL CALL
2004-08-27T15:12:35Z	prefetch run: NETSTUMBLER.EXE	—	tc-047
2004-08-27T15:46:13Z	UserAssist records execution of cain.exe	—	tc-214
2004-08-27T15:46:13Z	UserAssist records execution of cain25b45.exe	—	tc-214
2004-08-27T15:46:13Z	UserAssist records execution of ethereal-setup-0.10.6.exe	—	tc-214
2004-08-27T15:46:13Z	UserAssist records execution of ethereal.exe	—	tc-214
2004-08-27T15:46:13Z	UserAssist records execution of lookatlan.exe	—	tc-214
2004-08-27T15:46:13Z	UserAssist records execution of mirc.exe	—	tc-214
2004-08-27T15:46:13Z	UserAssist records execution of mirc612.exe	—	tc-214

LOOKATLAN.EXE-1F991DD9.PF

1 finding(s) — 1 confirmed, 0 inferred, 0 hypothesis · 1 events · 2004-08-26T15:06:14Z · source: LOOKATLAN.EXE-1F991DD9.pf

Other: T1046 [CONFIRMED] tc-038

The cited tool output meets a defined detection rule for this technique. Treat single-source signals as leads until corroborated across artifact classes.

TIME (UTC)	EVENT	ACCOUNT	TOOL CALL
2004-08-26T15:06:14Z	prefetch run: LOOKATLAN.EXE	—	tc-038
2004-08-27T15:46:13Z	UserAssist records execution of cain.exe	—	tc-214
2004-08-27T15:46:13Z	UserAssist records execution of cain25b45.exe	—	tc-214
2004-08-27T15:46:13Z	UserAssist records execution of ethereal-setup-0.10.6.exe	—	tc-214
2004-08-27T15:46:13Z	UserAssist records execution of ethereal.exe	—	tc-214
2004-08-27T15:46:13Z	UserAssist records execution of lookatlan.exe	—	tc-214
2004-08-27T15:46:13Z	UserAssist records execution of mirc.exe	—	tc-214
2004-08-27T15:46:13Z	UserAssist records execution of mirc612.exe	—	tc-214

MIRC.EXE-0661EC22.PF

1 finding(s) — 1 confirmed, 0 inferred, 0 hypothesis · 1 events · 2004-08-25T16:20:34Z · source: MIRC.EXE-0661EC22.pf

Command & Control: T1071.001 [CONFIRMED] tc-040

The cited tool output meets a defined detection rule for this technique. Treat single-source signals as leads until corroborated across artifact classes.

TIME (UTC)	EVENT	ACCOUNT	TOOL CALL
2004-08-25T16:20:34Z	prefetch run: MIRC.EXE	—	tc-040
2004-08-27T15:46:13Z	UserAssist records execution of cain.exe	—	tc-214
2004-08-27T15:46:13Z	UserAssist records execution of cain25b45.exe	—	tc-214
2004-08-27T15:46:13Z	UserAssist records execution of ethereal-setup-0.10.6.exe	—	tc-214
2004-08-27T15:46:13Z	UserAssist records execution of ethereal.exe	—	tc-214
2004-08-27T15:46:13Z	UserAssist records execution of lookatlan.exe	—	tc-214
2004-08-27T15:46:13Z	UserAssist records execution of mirc.exe	—	tc-214
2004-08-27T15:46:13Z	UserAssist records execution of mirc612.exe	—	tc-214

MIRC612.EXE-02791C37.PF

1 finding(s) — 1 confirmed, 0 inferred, 0 hypothesis · 1 events · 2004-08-20T15:09:46Z · source: MIRC612.EXE-02791C37.pf

Command & Control: T1071.001 [CONFIRMED] tc-041

The cited tool output meets a defined detection rule for this technique. Treat single-source signals as leads until corroborated across artifact classes.

TIME (UTC)	EVENT	ACCOUNT	TOOL CALL
2004-08-20T15:09:46Z	prefetch run: MIRC612.EXE	—	tc-041
2004-08-27T15:46:13Z	UserAssist records execution of cain.exe	—	tc-214
2004-08-27T15:46:13Z	UserAssist records execution of cain25b45.exe	—	tc-214
2004-08-27T15:46:13Z	UserAssist records execution of ethereal-setup-0.10.6.exe	—	tc-214
2004-08-27T15:46:13Z	UserAssist records execution of ethereal.exe	—	tc-214
2004-08-27T15:46:13Z	UserAssist records execution of lookatlan.exe	—	tc-214
2004-08-27T15:46:13Z	UserAssist records execution of mirc.exe	—	tc-214
2004-08-27T15:46:13Z	UserAssist records execution of mirc612.exe	—	tc-214

CAIN25B45.EXE-056F3A6E.PF

1 finding(s) — 1 confirmed, 0 inferred, 0 hypothesis · 1 events · 2004-08-20T15:05:52Z · source: CAIN25B45.EXE-056F3A6E.pf

Other: T1588.002 [CONFIRMED] tc-011

The cited tool output meets a defined detection rule for this technique. Treat single-source signals as leads until corroborated across artifact classes.

TIME (UTC)	EVENT	ACCOUNT	TOOL CALL
2004-08-20T15:05:52Z	prefetch run: CAIN25B45.EXE	—	tc-011
2004-08-27T15:46:13Z	UserAssist records execution of cain.exe	—	tc-214
2004-08-27T15:46:13Z	UserAssist records execution of cain25b45.exe	—	tc-214
2004-08-27T15:46:13Z	UserAssist records execution of ethereal-setup-0.10.6.exe	—	tc-214
2004-08-27T15:46:13Z	UserAssist records execution of ethereal.exe	—	tc-214
2004-08-27T15:46:13Z	UserAssist records execution of lookatlan.exe	—	tc-214
2004-08-27T15:46:13Z	UserAssist records execution of mirc.exe	—	tc-214
2004-08-27T15:46:13Z	UserAssist records execution of mirc612.exe	—	tc-214

NTUSER.DAT

6 finding(s) — 0 confirmed, 0 inferred, 6 hypothesis · 43 events · 2004-08-20T15:07:26Z → 2004-08-27T15:32:08Z · source: NTUSER.DAT

Other: T1074.001 [HYPOTHESIS] tc-179

The cited tool output meets a defined detection rule for this technique. Treat single-source signals as leads until corroborated across artifact classes.

Other: T1217 [HYPOTHESIS] tc-176

The cited tool output meets a defined detection rule for this technique. Treat single-source signals as leads until corroborated across artifact classes.

Other: T1217 [HYPOTHESIS] tc-176

The cited tool output meets a defined detection rule for this technique. Treat single-source signals as leads until corroborated across artifact classes.

Other: T1217 [HYPOTHESIS] tc-176

The cited tool output meets a defined detection rule for this technique. Treat single-source signals as leads until corroborated across artifact classes.

Other: T1217 [HYPOTHESIS] tc-176

The cited tool output meets a defined detection rule for this technique. Treat single-source signals as leads until corroborated across artifact classes.

Other: T1074.001 [HYPOTHESIS] tc-178

The cited tool output meets a defined detection rule for this technique. Treat single-source signals as leads until corroborated across artifact classes.

TIME (UTC)	EVENT	ACCOUNT	TOOL CALL
2004-08-20T15:07:26Z	registry key: Software\Microsoft\Windows\ShellNoRoam\BagMRU\0\1\0	—	tc-179
2004-08-20T15:07:26Z	registry key: Software\Microsoft\Windows\ShellNoRoam\BagMRU\0\1\0\0	—	tc-179
2004-08-20T15:07:26Z	registry key: Software\Microsoft\Windows\ShellNoRoam\BagMRU\0\1\0\0\0	—	tc-179
2004-08-20T15:07:26Z	registry key: Software\Microsoft\Windows\ShellNoRoam\BagMRU\0\1\0\0\0\0\0	—	tc-179
2004-08-20T15:09:56Z	registry key: Software\Microsoft\Windows\ShellNoRoam\BagMRU\0\1\0\0\0\1	—	tc-179
2004-08-20T15:10:06Z	registry key: Software\Microsoft\Windows\ShellNoRoam\BagMRU\0\1\0\0\0\0	—	tc-179
2004-08-20T15:11:11Z	registry key: Software\Microsoft\Windows\ShellNoRoam\BagMRU\0\1\1\0	—	tc-179
2004-08-20T15:19:00Z	registry key: Software\Microsoft\Windows\ShellNoRoam\BagMRU\0\0	—	tc-179

TARGET.LNK

3 finding(s) — 0 confirmed, 0 inferred, 3 hypothesis · 0 events · time not recorded · source: target.lnk

Other: T1074 [HYPOTHESIS] tc-055

The cited tool output meets a defined detection rule for this technique. Treat single-source signals as leads until corroborated across artifact classes.

Other: T1074 [HYPOTHESIS] tc-056

The cited tool output meets a defined detection rule for this technique. Treat single-source signals as leads until corroborated across artifact classes.

Other: T1074 [HYPOTHESIS] tc-057

The cited tool output meets a defined detection rule for this technique. Treat single-source signals as leads until corroborated across artifact classes.

NETSTUMBLERINSTALLER_0_4_0.EX-0BD9920C.PF

1 finding(s) — 0 confirmed, 0 inferred, 1 hypothesis · 1 events · 2004-08-27T15:12:11Z · source: NETSTUMBLERINSTALLER_0_4_0.EX-0BD9920C.pf

Other: T1046 [HYPOTHESIS] tc-048

The cited tool output meets a defined detection rule for this technique. Treat single-source signals as leads until corroborated across artifact classes.

TIME (UTC)	EVENT	ACCOUNT	TOOL CALL
2004-08-27T15:12:11Z	prefetch run: NETSTUMBLERINSTALLER_0_4_0.EX	—	tc-048

SAM

1 finding(s) — 0 confirmed, 0 inferred, 1 hypothesis · 6 events · 2004-08-19T16:59:24Z → 2004-08-19T23:03:54Z · source: SAM

Persistence: T1136.001 [HYPOTHESIS] tc-206

The cited tool output meets a defined detection rule for this technique. Treat single-source signals as leads until corroborated across artifact classes.

TIME (UTC)	EVENT	ACCOUNT	TOOL CALL
2004-08-19T16:59:24Z	registry key: SAM\Domains\Account\Users\Names\Administrator	—	tc-206
2004-08-19T16:59:24Z	registry key: SAM\Domains\Account\Users\Names\Guest	—	tc-206
2004-08-19T22:28:24Z	registry key: SAM\Domains\Account\Users\Names\HelpAssistant	—	tc-206
2004-08-19T22:35:19Z	registry key: SAM\Domains\Account\Users\Names\SUPPORT_388945a0	—	tc-206
2004-08-19T23:03:54Z	registry key: SAM\Domains\Account\Users\Names	—	tc-206
2004-08-19T23:03:54Z	registry key: SAM\Domains\Account\Users\Names\Mr. Evil	—	tc-206

$MFT

1 finding(s) — 0 confirmed, 0 inferred, 1 hypothesis · 497 events · 1601-01-01T00:00:00Z → 2004-08-27T15:08:33Z · source: $MFT

Other: T1588.002 [HYPOTHESIS] tc-004

The cited tool output meets a defined detection rule for this technique. Treat single-source signals as leads until corroborated across artifact classes.

TIME (UTC)	EVENT	ACCOUNT	TOOL CALL
1601-01-01T00:00:00Z	mft entry: $Secure	—	tc-004
1999-04-23T22:22:00Z	mft entry: WIN98/BASE4.CAB	—	tc-004
1999-04-23T22:22:00Z	mft entry: WIN98/BASE5.CAB	—	tc-004
1999-04-23T22:22:00Z	mft entry: WIN98/BASE6.CAB	—	tc-004
1999-04-23T22:22:00Z	mft entry: WIN98/CATALOG3.CAB	—	tc-004
1999-04-23T22:22:00Z	mft entry: WIN98/CHL99.CAB	—	tc-004
1999-04-23T22:22:00Z	mft entry: WIN98/DELTEMP.COM	—	tc-004
1999-04-23T22:22:00Z	mft entry: WIN98/DOSSETUP.BIN	—	tc-004

ANONYYMIZER.LNK

1 finding(s) — 0 confirmed, 0 inferred, 1 hypothesis · 0 events · time not recorded · source: Anonyymizer.lnk

Other: T1074 [HYPOTHESIS] tc-058

The cited tool output meets a defined detection rule for this technique. Treat single-source signals as leads until corroborated across artifact classes.

CHANNELS (2).LNK

1 finding(s) — 0 confirmed, 0 inferred, 1 hypothesis · 0 events · time not recorded · source: channels (2).lnk

Other: T1074 [HYPOTHESIS] tc-059

The cited tool output meets a defined detection rule for this technique. Treat single-source signals as leads until corroborated across artifact classes.

CHANNELS.LNK

1 finding(s) — 0 confirmed, 0 inferred, 1 hypothesis · 0 events · time not recorded · source: channels.lnk

Other: T1074 [HYPOTHESIS] tc-060

The cited tool output meets a defined detection rule for this technique. Treat single-source signals as leads until corroborated across artifact classes.

GHOSTWARE.LNK

1 finding(s) — 0 confirmed, 0 inferred, 1 hypothesis · 0 events · time not recorded · source: GhostWare.lnk

Other: T1074 [HYPOTHESIS] tc-061

The cited tool output meets a defined detection rule for this technique. Treat single-source signals as leads until corroborated across artifact classes.

KEYS.LNK

1 finding(s) — 0 confirmed, 0 inferred, 1 hypothesis · 0 events · time not recorded · source: keys.lnk

Other: T1074 [HYPOTHESIS] tc-062

The cited tool output meets a defined detection rule for this technique. Treat single-source signals as leads until corroborated across artifact classes.

TEMP ON M1200 (4.12.220.254).LNK

1 finding(s) — 0 confirmed, 0 inferred, 1 hypothesis · 0 events · time not recorded · source: Temp on m1200 (4.12.220.254).lnk

Other: T1074 [HYPOTHESIS] tc-063

The cited tool output meets a defined detection rule for this technique. Treat single-source signals as leads until corroborated across artifact classes.

INFO2

1 finding(s) — 0 confirmed, 0 inferred, 1 hypothesis · 0 events · time not recorded · source: INFO2

Defense Evasion: T1070.004 [HYPOTHESIS] tc-135

The cited tool output meets a defined detection rule for this technique. Treat single-source signals as leads until corroborated across artifact classes.

RECOMMENDATIONS

- Collect Security, Sysmon, and PowerShell Operational EVTX and rerun EVTX/Hayabusa analysis.
- Use the disk artifact summary to pivot between Prefetch, Registry, MFT, USN, EVTX, and YARA-target rows without upgrading single-source execution claims.
- Acquire DNS, proxy, firewall, NetFlow, or PCAP telemetry to test C2 and exfiltration hypotheses.

LIMITATIONS

What the supplied evidence cannot establish, and how to resolve it.

- Who operated the activity — this report does not assert attribution; naming an account reflects a record field, not the human behind it.
- Whether the wider environment is affected — this run examined the supplied evidence only.
- Undetermined: command-and-control and exfiltration cannot be assessed without network data. To resolve: collect DNS, proxy, firewall, or NetFlow logs, or a PCAP.
- Undetermined: injected code and hidden processes cannot be examined without a memory image. To resolve: capture RAM and run the volatility process and injection plugins.
- Undetermined: logon, process-creation, and PowerShell activity cannot be reviewed without event logs. To resolve: collect the Security, Sysmon/Operational, and PowerShell/Operational logs.
- A tool did not complete (tool error); raw output is in the run audit log. Resolve the tool’s prerequisites and re-run.

TECHNICAL REPORT

The sections below are the full analyst-grade record: every finding with its `tool_call_id` and confidence, the complete event timeline, the entity rollup and indicators, coverage matrices, triage, sources, and the reproducibility and chain-of-custody appendices.

CASE SUMMARY

- **Total merged findings:** 27
- **By confidence:**
  - CONFIRMED: 8
  - INFERRED: 0
  - HYPOTHESIS: 19
- **Contradictions surfaced (Pool A vs Pool B):** 8
- **SOUL.md correlator:** 27 kept, 0 downgraded

COVERAGE MANIFEST

If no parser/tool extracts an artifact class, VERDICT cannot reason over it. This manifest records that boundary explicitly.

This table is the explicit anti-overclaim record for the run. `not_supplied`, `unsupported`, `failed`, and `partial` rows are scope gaps, not clean findings.

- Artifact classes recorded: `11`
- Attempted: `7`; parsed: `7`; failed: `1`
- Unsupported: `0`; not supplied: `4`
- ATT&CK blind spots: `5`
- Status counts: `{'not_supplied': 4, 'parsed': 6, 'partial': 1}`

ARTIFACT CLASS	STATUS	AVAILABLE	ATTEMPTED	PARSED	FAILED	UNSUPPORTED	NOT SUPPLIED	PARSE ERRORS	RECORDS SEEN	ROWS RETURNED	TOOLS
custody	`parsed`	yes	yes	yes	no	no	no				`case_open`
disk/filesystem	`parsed`	yes	yes	yes	no	no	no		9		`ez_parse`, `ez_par`

ARTIFACT CLASS	STATUS	AVAILABLE	ATTEMPTED	PARSED	FAILED	UNSUPPORTED	NOT SUPPLIED	PARSE ERRORS	RECORDS SEEN	ROWS RETURNED	TOOLS
											`se,` `ez_par` `se,`

<table>
<tr><th>ARTIFACT CLASS</th><th>STATUS</th><th>AVAILABLE</th><th>ATTEMPTED</th><th>PARSED</th><th>FAILED</th><th>UNSUPPORTED</th><th>NOT SUPPLIED</th><th>PARSE ERRORS</th><th>RECORDS SEEN</th><th>ROWS RETURNED</th><th>TOOLS</th></tr>
<tr><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td>ez_parse, ez_parse, ez_parse, ez_parse, ez_parse, ez_parse, ez_parse, ez_parse, ez_parse, ez_parse, ez_parse, ez_parse, ez_parse, ez_parse, ez_parse, ez_parse, ez_parse, ez_parse</td></tr>
<tr><td>evtx</td><td>not_supplied</td><td>no</td><td>no</td><td>no</td><td>no</td><td>no</td><td>yes</td><td></td><td></td><td></td><td>| | memory | `not_supplied` | no | no | no | no | no | yes | | | |</td></tr>
<tr><td>mft</td><td>parsed</td><td>yes</td><td>yes</td><td>yes</td><td>no</td><td>no</td><td>no</td><td></td><td>5000</td><td>5000</td><td>mft_timeline</td></tr>
<tr><td>network</td><td>not_supplied</td><td>no</td><td>no</td><td>no</td><td>no</td><td>no</td><td>yes</td><td></td><td></td><td></td><td>| | prefetch | `parsed` | yes | yes | yes | no | no | no | | | | `prefetch_parse, prefetch_parse, prefetch_parse, prefetch_parse, prefet</td></tr>
</table>

ARTIFA CT CLASS	STATU S	AVAILABLE	ATTEMPTED	PARSED	FAILED	UNSUPPORTED	NOT SUPPLIED	PARSE ERRORS	RECORDS SEEN	ROWS RETURNED	TOOLS
											`ch_parse`, `prefetch_parse`, `prefet`

ARTIFACT CLASS	STATUS	AVAILABLE	ATTEMPTED	PARSED	FAILED	UNSUPPORTED	NOT SUPPLIED	PARSE ERRORS	RECORDS SEEN	ROWS RETURNED	TOOLS
											ch_parse, prefetch_parse, prefetch_parse, prefetch_parse, prefetch_parse, prefetch_parse, prefetch_parse, prefetch_parse, prefetch_parse, prefetch_parse, prefetch_parse, prefetch_parse, prefetch_parse, prefetch_parse, prefetch_parse, prefetch_parse, prefetch_parse, prefetch_parse` | | registry | `parsed` | yes | yes | yes | no | no | no | | | | `registry_query, registry_query, registry_query, registry_query,

ARTIFACT CLASS	STATUS	AVAILABLE	ATTEMPTED	PARSED	FAILED	UNSUPPORTED	NOT SUPPLIED	PARSE ERRORS	RECORDS SEEN	ROWS RETURNED	TOOLS
											`registry_query`, `registry_query`,

ARTIFACT CLASS	STATUS	AVAILABLE	ATTEMPTED	PARSED	FAILED	UNSUPPORTED	NOT SUPPLIED	PARSE ERRORS	RECORDS SEEN	ROWS RETURNED	TOOLS
											`registry_query`, `registry_query`,

<table>
<tr><th>ARTIFACT CLASS</th><th>STATUS</th><th>AVAILABLE</th><th>ATTEMPTED</th><th>PARSED</th><th>FAILED</th><th>UNSUPPORTED</th><th>NOT SUPPLIED</th><th>PARSE ERRORS</th><th>RECORDS SEEN</th><th>ROWS RETURNED</th><th>TOOLS</th></tr>
<tr><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td>registry_query, registry_query, registry_query, registry_query, registry_query, registry_query, registry_query` | | timeline | `partial` | yes | yes | yes | yes | no | no | | 4 | | `plaso_parse, plaso_parse, plaso_parse, plaso_parse, plaso_parse, plaso_parse, plaso_parse, plaso_parse, plaso_parse, plaso_parse, plaso_parse, plaso_parse, plaso_parse, plaso_parse, plaso_parse, plaso_parse, plaso_parse` | | unknown_tool_output | `parsed` | yes | yes | yes | no | no |</td></tr>
</table>

<table>
<thead>
<tr><th>ARTIFACT CLASS</th><th>STATUS</th><th>AVAILABLE</th><th>ATTEMPTED</th><th>PARSED</th><th>FAILED</th><th>UNSUPPORTED</th><th>NOT SUPPLIED</th><th>PARSE ERRORS</th><th>RECORDS SEEN</th><th>ROWS RETURNED</th><th>TOOLS</th></tr>
</thead>
<tbody>
<tr><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td>no | | 276 | | | `disk_mount, disk_extract_artifacts, disk_unmount` | | velociraptor | `not_supplied` | no | no | no | no | no | yes | | | |</td></tr>
</tbody>
</table>

DETAILED FINDINGS

FINDING 1 — CONFIDENCE: HYPOTHESIS, POOL: A, MITRE: T1588.002, REPLAY: EXACT_MATCH (MATCH)

hypothesis: Hacking-tool artifacts present on disk as downloaded applications, recovered from the MFT: Program Files/Anonymizer (created 2004-08-20T15:05:06Z). The filesystem records (MFT) show these tool files in Program Files / on the user's Desktop with creation timestamps clustered around the incident window; they are downloaded applications, not operating-system components. INFERRED from two tool-backed facts per file: the artifact's existence (MFT) and its name matching a known-tool heuristic (anonymizer). Corroborates Prefetch observations for the same toolset; file presence itself remains a filesystem fact.

- `tool_call_id`: `tc-004`
- artifact: `/home/assessor/.findevil/cases/bb2953cd-bdd9-4251-a493-70c97663f892/extracted/disk/disk-extract-affa2c27-51ae-4505-912a-27a7ec2a93a8/mft/$MFT`
- confidence: Hypothesis — a single-source triage lead; corroborate before any response action.

FINDING 2 — CONFIDENCE: HYPOTHESIS, POOL: A, MITRE: T1217, REPLAY: EXACT_MATCH (MATCH)

hypothesis: User recently opened-file MRU records 'C:\Documents and Settings\Mr. Evil\Desktop\lalsetup250.exe' (Software\Microsoft\Windows\CurrentVersion\Explorer\ComDlg32\OpenSaveMRU*, registry_query, last_write 2004-08-27T15:18:05Z). INFERRED user activity: the MRU entry's existence is tool-backed and reflects deliberate user action. It records intent/recency only.

- `tool_call_id`: `tc-176`
- artifact: `/home/assessor/.findevil/cases/bb2953cd-bdd9-4251-a493-70c97663f892/extracted/disk/disk-extract-affa2c27-51ae-4505-912a-27a7ec2a93a8/registry/Documents and Settings/Mr. Evil/NTUSER.DAT`
- confidence: Hypothesis — a single-source triage lead; corroborate before any response action.

FINDING 3 — CONFIDENCE: HYPOTHESIS, POOL: A, MITRE: T1217, REPLAY: EXACT_MATCH (MATCH)

hypothesis: User recently opened-file MRU records 'C:\Documents and Settings\Mr. Evil\Desktop\netstumblerinstaller_0_4_0.exe' (Software\Microsoft\Windows\CurrentVersion\Explorer\ComDlg32\OpenSaveMRU*, registry_query, last_write 2004-08-27T15:18:05Z). INFERRED user activity: the MRU entry's existence is tool-backed and reflects deliberate user action. It records intent/recency only.

- `tool_call_id`: `tc-176`
- artifact: `/home/assessor/.findevil/cases/bb2953cd-bdd9-4251-a493-70c97663f892/extracted/disk/disk-extract-affa2c27-51ae-4505-912a-27a7ec2a93a8/registry/Documents and Settings/Mr. Evil/NTUSER.DAT`
- confidence: Hypothesis — a single-source triage lead; corroborate before any response action.

FINDING 4 — CONFIDENCE: HYPOTHESIS, POOL: A, MITRE: T1217, REPLAY: EXACT_MATCH (MATCH)

hypothesis: User recently opened-file MRU records 'C:\Documents and Settings\Mr. Evil\Desktop\ethereal-setup-0.10.6.exe' (Software\Microsoft\Windows\CurrentVersion\Explorer\ComDlg32\OpenSaveMRU*, registry_query, last_write 2004-08-27T15:18:05Z). INFERRED user activity: the MRU entry's existence is tool-backed and reflects deliberate user action. It records intent/recency only.

- `tool_call_id`: `tc-176`
- artifact: `/home/assessor/.findevil/cases/bb2953cd-bdd9-4251-a493-70c97663f892/extracted/disk/disk-extract-affa2c27-51ae-4505-912a-27a7ec2a93a8/registry/Documents and Settings/Mr. Evil/NTUSER.DAT`
- confidence: Hypothesis — a single-source triage lead; corroborate before any response action.

FINDING 5 — CONFIDENCE: **HYPOTHESIS**, POOL: A, MITRE: T1217, REPLAY: EXACT_MATCH (MATCH)

hypothesis: User recently opened-file MRU records 'C:\Documents and Settings\Mr. Evil\Desktop\WinPcap_3_01_a.exe' (Software\Microsoft\Windows\CurrentVersion\Explorer\ComDlg32\OpenSaveMRU*, registry_query, last_write 2004-08-27T15:18:05Z). **INFERRED** user activity: the MRU entry's existence is tool-backed and reflects deliberate user action. It records intent/recency only.

- `tool_call_id`: `tc-176`
- artifact: `/home/assessor/.findevil/cases/bb2953cd-bdd9-4251-a493-70c97663f892/extracted/disk/disk-extract-affa2c27-51ae-4505-912a-27a7ec2a93a8/registry/Documents and Settings/Mr. Evil/NTUSER.DAT`
- confidence: Hypothesis — a single-source triage lead; corroborate before any response action.

FINDING 6 — CONFIDENCE: **HYPOTHESIS**, POOL: A, MITRE: T1136.001, REPLAY: EXACT_MATCH (MATCH)

hypothesis: User account 'Mr. Evil' with suspicious naming was created on this system: it is recorded in the SAM (Security Account Manager) hive (SAM\Domains\Account\Users\Names\Mr. Evil; the Names subkey last_write 2004-08-19T23:03:54Z approximates the account-creation time). **INFERRED** from two tool-backed facts: the account's existence (registry_query) and its name matching the suspicious-naming heuristic. Whether it holds elevated privileges is NOT yet claimed — enumerate group membership (Administrators) and logon artifacts to corroborate.

- `tool_call_id`: `tc-206`
- artifact: `/home/assessor/.findevil/cases/bb2953cd-bdd9-4251-a493-70c97663f892/extracted/disk/disk-extract-affa2c27-51ae-4505-912a-27a7ec2a93a8/registry/WINDOWS/system32/config/SAM`
- confidence: Hypothesis — a single-source triage lead; corroborate before any response action.

FINDING 7 — CONFIDENCE: **CONFIRMED**, POOL: B, MITRE: T1588.002, REPLAY: EXACT_MATCH (MATCH)

cain.exe executed on this host: Windows Prefetch records its execution and the UserAssist key (per-user GUI execution) records the same binary. Two independent artifact classes (prefetch + registry/UserAssist) corroborate execution.

- `tool_call_id`: `tc-010`
- artifact: `/home/assessor/.findevil/cases/bb2953cd-bdd9-4251-a493-70c97663f892/extracted/disk/disk-extract-affa2c27-51ae-4505-912a-27a7ec2a93a8/prefetch/WINDOWS/Prefetch/CAIN.EXE-23D61279.pf`
- confidence: Confirmed — the cited tool output is reproducible; this does not imply attribution or complete scope.

FINDING 8 — CONFIDENCE: **CONFIRMED**, POOL: B, MITRE: T1588.002, REPLAY: EXACT_MATCH (MATCH)

cain25b45.exe executed on this host: Windows Prefetch records its execution and the UserAssist key (per-user GUI execution) records the same binary. Two independent artifact classes (prefetch + registry/UserAssist) corroborate execution.

- `tool_call_id`: `tc-011`
- artifact: `/home/assessor/.findevil/cases/bb2953cd-bdd9-4251-a493-70c97663f892/extracted/disk/disk-extract-affa2c27-51ae-4505-912a-27a7ec2a93a8/prefetch/WINDOWS/Prefetch/CAIN25B45.EXE-056F3A6E.pf`
- confidence: Confirmed — the cited tool output is reproducible; this does not imply attribution or complete scope.

FINDING 9 — CONFIDENCE: **CONFIRMED**, POOL: B, MITRE: T1040, REPLAY: EXACT_MATCH (MATCH)

ethereal-setup-0.10.6.exe executed on this host: Windows Prefetch records its execution and the UserAssist key (per-user GUI execution) records the same binary. Two independent artifact classes (prefetch + registry/UserAssist) corroborate execution.

- `tool_call_id`: `tc-020`
- artifact: `/home/assessor/.findevil/cases/bb2953cd-bdd9-4251-a493-70c97663f892/extracted/disk/disk-extract-affa2c27-51ae-4505-912a-27a7ec2a93a8/prefetch/WINDOWS/Prefetch/ETHEREAL-SETUP-0.10.6.EXE-1D932600.pf`
- confidence: Confirmed — the cited tool output is reproducible; this does not imply attribution or complete scope.

FINDING 10 — CONFIDENCE: **CONFIRMED**, POOL: B, MITRE: T1040, REPLAY: EXACT_MATCH (MATCH)

ethereal.exe executed on this host: Windows Prefetch records its execution and the UserAssist key (per-user GUI execution) records the same binary. Two independent artifact classes (prefetch + registry/UserAssist) corroborate execution.

- `tool_call_id`: `tc-021`
- artifact: `/home/assessor/.findevil/cases/bb2953cd-bdd9-4251-a493-70c97663f892/extracted/disk/disk-extract-affa2c27-51ae-4505-912a-27a7ec2a93a8/prefetch/WINDOWS/Prefetch/ETHEREAL.EXE-1C148EEF.pf`
- confidence: Confirmed — the cited tool output is reproducible; this does not imply attribution or complete scope.

FINDING 11 — CONFIDENCE: **CONFIRMED**, POOL: B, MITRE: T1046, REPLAY: EXACT_MATCH (MATCH)

lookatlan.exe executed on this host: Windows Prefetch records its execution and the UserAssist key (per-user GUI execution) records the same binary. Two independent artifact classes (prefetch + registry/UserAssist) corroborate execution.

- `tool_call_id`: `tc-038`
- artifact: `/home/assessor/.findevil/cases/bb2953cd-bdd9-4251-a493-70c97663f892/extracted/disk/disk-extract-affa2c27-51ae-4505-912a-27a7ec2a93a8/prefetch/WINDOWS/Prefetch/LOOKATLAN.EXE-1F991DD9.pf`
- confidence: Confirmed — the cited tool output is reproducible; this does not imply attribution or complete scope.

FINDING 12 — CONFIDENCE: **CONFIRMED**, POOL: B, MITRE: T1071.001, REPLAY: EXACT_MATCH (MATCH)

mirc.exe executed on this host: Windows Prefetch records its execution and the UserAssist key (per-user GUI execution) records the same binary. Two independent artifact classes (prefetch + registry/UserAssist) corroborate execution.

- `tool_call_id`: `tc-040`
- artifact: `/home/assessor/.findevil/cases/bb2953cd-bdd9-4251-a493-70c97663f892/extracted/disk/disk-extract-affa2c27-51ae-4505-912a-27a7ec2a93a8/prefetch/WINDOWS/Prefetch/MIRC.EXE-0661EC22.pf`
- confidence: Confirmed — the cited tool output is reproducible; this does not imply attribution or complete scope.

FINDING 13 — CONFIDENCE: CONFIRMED, POOL: B, MITRE: T1071.001, REPLAY: EXACT_MATCH (MATCH)

mirc612.exe executed on this host: Windows Prefetch records its execution and the UserAssist key (per-user GUI execution) records the same binary. Two independent artifact classes (prefetch + registry/UserAssist) corroborate execution.

- `tool_call_id`: `tc-041`
- artifact: `/home/assessor/.findevil/cases/bb2953cd-bdd9-4251-a493-70c97663f892/extracted/disk/disk-extract-affa2c27-51ae-4505-912a-27a7ec2a93a8/prefetch/WINDOWS/Prefetch/MIRC612.EXE-02791C37.pf`
- confidence: Confirmed — the cited tool output is reproducible; this does not imply attribution or complete scope.

FINDING 14 — CONFIDENCE: CONFIRMED, POOL: B, MITRE: T1046, REPLAY: EXACT_MATCH (MATCH)

netstumbler.exe executed on this host: Windows Prefetch records its execution and the UserAssist key (per-user GUI execution) records the same binary. Two independent artifact classes (prefetch + registry/UserAssist) corroborate execution.

- `tool_call_id`: `tc-047`
- artifact: `/home/assessor/.findevil/cases/bb2953cd-bdd9-4251-a493-70c97663f892/extracted/disk/disk-extract-affa2c27-51ae-4505-912a-27a7ec2a93a8/prefetch/WINDOWS/Prefetch/NETSTUMBLER.EXE-0BFEE568.pf`
- confidence: Confirmed — the cited tool output is reproducible; this does not imply attribution or complete scope.

FINDING 15 — CONFIDENCE: HYPOTHESIS, POOL: B, MITRE: T1046, REPLAY: EXACT_MATCH (MATCH)

hypothesis: Windows Prefetch contains NETSTUMBLERINSTALLER_0_4_0.EX with run_count=1; NetStumbler wireless discovery tool is a NIST Hacking Case triage lead. Treat this as a disk-artifact lead that needs corroboration before any standalone activity claim.

- `tool_call_id`: `tc-048`
- artifact: `/home/assessor/.findevil/cases/bb2953cd-bdd9-4251-a493-70c97663f892/extracted/disk/disk-extract-affa2c27-51ae-4505-912a-27a7ec2a93a8/prefetch/WINDOWS/Prefetch/NETSTUMBLERINSTALLER_0_4_0.EX-0BD9920C.pf`
- confidence: Hypothesis — a single-source triage lead; corroborate before any response action.

FINDING 16 — CONFIDENCE: HYPOTHESIS, POOL: B, MITRE: T1074, REPLAY: EXACT_MATCH (MATCH)

hypothesis: LNK shortcut artifact references removable media activity: /home/assessor/.findevil/cases/bb2953cd-bdd9-4251-a493-70c97663f892/extracted/disk/disk-extract-affa2c27-51ae-4505-912a-27a7ec2a93a8/lnk/Documents and Settings/Mr. Evil/NetHood/CD Drive (F) on Andrews-1/target.lnk. LECmd was unavailable for at least one shortcut, so path-only Recent/NetHood context is used and volume serial is not claimed for those rows. Treat this as a shortcut/removable-media staging lead only; if the source path is a Recent folder, use it as a recent activity pivot. Corroborate with filesystem, registry, event-log, or network evidence before asserting user activity.

- `tool_call_id`: `tc-055`
- artifact: `/home/assessor/.findevil/cases/bb2953cd-bdd9-4251-a493-70c97663f892/extracted/disk/disk-extract-affa2c27-51ae-4505-912a-27a7ec2a93a8/lnk/Documents and Settings/Mr. Evil/NetHood/CD Drive \(F\) on Andrews-1/target.lnk`
- confidence: Hypothesis — a single-source triage lead; corroborate before any response action.

FINDING 17 — CONFIDENCE: HYPOTHESIS, POOL: B, MITRE: T1074, REPLAY: EXACT_MATCH (MATCH)

hypothesis: LNK shortcut artifact references removable media activity: /home/assessor/.findevil/cases/bb2953cd-bdd9-4251-a493-70c97663f892/extracted/disk/disk-extract-affa2c27-51ae-4505-912a-27a7ec2a93a8/lnk/Documents and Settings/Mr. Evil/NetHood/Temp on m1200 (4.12.220.254)/target.lnk. LECmd was unavailable for at least one shortcut, so path-only Recent/NetHood context is used and volume serial is not claimed for those rows. Treat this as a shortcut/removable-media staging lead only; if the source path is a Recent folder, use it as a recent activity pivot. Corroborate with filesystem, registry, event-log, or network evidence before asserting user activity.

- `tool_call_id`: `tc-056`
- artifact: `/home/assessor/.findevil/cases/bb2953cd-bdd9-4251-a493-70c97663f892/extracted/disk/disk-extract-affa2c27-51ae-4505-912a-27a7ec2a93a8/lnk/Documents and Settings/Mr. Evil/NetHood/Temp on m1200 \(4.12.220.254\)/target.lnk`
- confidence: Hypothesis — a single-source triage lead; corroborate before any response action.

FINDING 18 — CONFIDENCE: HYPOTHESIS, POOL: B, MITRE: T1074, REPLAY: EXACT_MATCH (MATCH)

hypothesis: LNK shortcut artifact references removable media activity: /home/assessor/.findevil/cases/bb2953cd-bdd9-4251-a493-70c97663f892/extracted/disk/disk-extract-affa2c27-51ae-4505-912a-27a7ec2a93a8/lnk/Documents and Settings/Mr. Evil/NetHood/Temp on N-1a9odn6zxk4lq/target.lnk. LECmd was unavailable for at least one shortcut, so path-only Recent/NetHood context is used and volume serial is not claimed for those rows. Treat this as a shortcut/removable-media staging lead only; if the source path is a Recent folder, use it as a recent activity pivot. Corroborate with filesystem, registry, event-log, or network evidence before asserting user activity.

- `tool_call_id`: `tc-057`
- artifact: `/home/assessor/.findevil/cases/bb2953cd-bdd9-4251-a493-70c97663f892/extracted/disk/disk-extract-affa2c27-51ae-4505-912a-27a7ec2a93a8/lnk/Documents and Settings/Mr. Evil/NetHood/Temp on N-1a9odn6zxk4lq/target.lnk`
- confidence: Hypothesis — a single-source triage lead; corroborate before any response action.

FINDING 19 — CONFIDENCE: **HYPOTHESIS**, POOL: B, MITRE: T1074, REPLAY: EXACT_MATCH (MATCH)

hypothesis: LNK shortcut artifact references removable media activity: /home/assessor/.findevil/cases/bb2953cd-bdd9-4251-a493-70c97663f892/extracted/disk/disk-extract-affa2c27-51ae-4505-912a-27a7ec2a93a8/lnk/Documents and Settings/Mr. Evil/Recent/Anonyymizer.lnk. LECmd was unavailable for at least one shortcut, so path-only Recent/NetHood context is used and volume serial is not claimed for those rows. Treat this as a shortcut/removable-media staging lead only; if the source path is a Recent folder, use it as a recent activity pivot. Corroborate with filesystem, registry, event-log, or network evidence before asserting user activity.

- `tool_call_id`: `tc-058`
- artifact: `/home/assessor/.findevil/cases/bb2953cd-bdd9-4251-a493-70c97663f892/extracted/disk/disk-extract-affa2c27-51ae-4505-912a-27a7ec2a93a8/lnk/Documents and Settings/Mr. Evil/Recent/Anonyymizer.lnk`
- confidence: Hypothesis — a single-source triage lead; corroborate before any response action.

FINDING 20 — CONFIDENCE: **HYPOTHESIS**, POOL: B, MITRE: T1074, REPLAY: EXACT_MATCH (MATCH)

hypothesis: LNK shortcut artifact references removable media activity: /home/assessor/.findevil/cases/bb2953cd-bdd9-4251-a493-70c97663f892/extracted/disk/disk-extract-affa2c27-51ae-4505-912a-27a7ec2a93a8/lnk/Documents and Settings/Mr. Evil/Recent/channels (2).lnk. LECmd was unavailable for at least one shortcut, so path-only Recent/NetHood context is used and volume serial is not claimed for those rows. Treat this as a shortcut/removable-media staging lead only; if the source path is a Recent folder, use it as a recent activity pivot. Corroborate with filesystem, registry, event-log, or network evidence before asserting user activity.

- `tool_call_id`: `tc-059`
- artifact: `/home/assessor/.findevil/cases/bb2953cd-bdd9-4251-a493-70c97663f892/extracted/disk/disk-extract-affa2c27-51ae-4505-912a-27a7ec2a93a8/lnk/Documents and Settings/Mr. Evil/Recent/channels \(2\).lnk`
- confidence: Hypothesis — a single-source triage lead; corroborate before any response action.

FINDING 21 — CONFIDENCE: **HYPOTHESIS**, POOL: B, MITRE: T1074, REPLAY: EXACT_MATCH (MATCH)

hypothesis: LNK shortcut artifact references removable media activity: /home/assessor/.findevil/cases/bb2953cd-bdd9-4251-a493-70c97663f892/extracted/disk/disk-extract-affa2c27-51ae-4505-912a-27a7ec2a93a8/lnk/Documents and Settings/Mr. Evil/Recent/channels.lnk. LECmd was unavailable for at least one shortcut, so path-only Recent/NetHood context is used and volume serial is not claimed for those rows. Treat this as a shortcut/removable-media staging lead only; if the source path is a Recent folder, use it as a recent activity pivot. Corroborate with filesystem, registry, event-log, or network evidence before asserting user activity.

- `tool_call_id`: `tc-060`
- artifact: `/home/assessor/.findevil/cases/bb2953cd-bdd9-4251-a493-70c97663f892/extracted/disk/disk-extract-affa2c27-51ae-4505-912a-27a7ec2a93a8/lnk/Documents and Settings/Mr. Evil/Recent/channels.lnk`
- confidence: Hypothesis — a single-source triage lead; corroborate before any response action.

FINDING 22 — CONFIDENCE: **HYPOTHESIS**, POOL: B, MITRE: T1074, REPLAY: EXACT_MATCH (MATCH)

hypothesis: LNK shortcut artifact references removable media activity: /home/assessor/.findevil/cases/bb2953cd-bdd9-4251-a493-70c97663f892/extracted/disk/disk-extract-affa2c27-51ae-4505-912a-27a7ec2a93a8/lnk/Documents and Settings/Mr. Evil/Recent/GhostWare.lnk. LECmd was unavailable for at least one shortcut, so path-only Recent/NetHood context is used and volume serial is not claimed for those rows. Treat this as a shortcut/removable-media staging lead only; if the source path is a Recent folder, use it as a recent activity pivot. Corroborate with filesystem, registry, event-log, or network evidence before asserting user activity.

- `tool_call_id`: `tc-061`
- artifact: `/home/assessor/.findevil/cases/bb2953cd-bdd9-4251-a493-70c97663f892/extracted/disk/disk-extract-affa2c27-51ae-4505-912a-27a7ec2a93a8/lnk/Documents and Settings/Mr. Evil/Recent/GhostWare.lnk`
- confidence: Hypothesis — a single-source triage lead; corroborate before any response action.

FINDING 23 — CONFIDENCE: **HYPOTHESIS**, POOL: B, MITRE: T1074, REPLAY: EXACT_MATCH (MATCH)

hypothesis: LNK shortcut artifact references removable media activity: /home/assessor/.findevil/cases/bb2953cd-bdd9-4251-a493-70c97663f892/extracted/disk/disk-extract-affa2c27-51ae-4505-912a-27a7ec2a93a8/lnk/Documents and Settings/Mr. Evil/Recent/keys.lnk. LECmd was unavailable for at least one shortcut, so path-only Recent/NetHood context is used and volume serial is not claimed for those rows. Treat this as a shortcut/removable-media staging lead only; if the source path is a Recent folder, use it as a recent activity pivot. Corroborate with filesystem, registry, event-log, or network evidence before asserting user activity.

- `tool_call_id`: `tc-062`
- artifact: `/home/assessor/.findevil/cases/bb2953cd-bdd9-4251-a493-70c97663f892/extracted/disk/disk-extract-affa2c27-51ae-4505-912a-27a7ec2a93a8/lnk/Documents and Settings/Mr. Evil/Recent/keys.lnk`
- confidence: Hypothesis — a single-source triage lead; corroborate before any response action.

FINDING 24 — CONFIDENCE: **HYPOTHESIS**, POOL: B, MITRE: T1074, REPLAY: EXACT_MATCH (MATCH)

hypothesis: LNK shortcut artifact references removable media activity: /home/assessor/.findevil/cases/bb2953cd-bdd9-4251-a493-70c97663f892/extracted/disk/disk-extract-affa2c27-51ae-4505-912a-27a7ec2a93a8/lnk/Documents and Settings/Mr. Evil/Recent/Temp on m1200 (4.12.220.254).lnk. LECmd was unavailable for at least one shortcut, so path-only Recent/NetHood context is used and volume serial is not claimed for those rows. Treat this as a shortcut/removable-media staging lead only; if the source path is a Recent folder, use it as a recent activity pivot. Corroborate with filesystem, registry, event-log, or network evidence before asserting user activity.

- `tool_call_id`: `tc-063`
- artifact: `/home/assessor/.findevil/cases/bb2953cd-bdd9-4251-a493-70c97663f892/extracted/disk/disk-extract-affa2c27-51ae-4505-912a-27a7ec2a93a8/lnk/Documents and Settings/Mr. Evil/Recent/Temp on m1200 \(4.12.220.254\).lnk`
- confidence: Hypothesis — a single-source triage lead; corroborate before any response action.

FINDING 25 — CONFIDENCE: HYPOTHESIS, POOL: B, MITRE: T1070.004, REPLAY: EXACT_MATCH (MATCH)

hypothesis: Recycle Bin recycle_bin_info2 deleted-item artifact records a deleted staging/tool artifact: C:\Documents and Settings\Mr. Evil\Desktop\WinPcap_3_01_a.exe; C:\Documents and Settings\Mr. Evil\Desktop\ethereal-setup-0.10.6.exe; C:\Documents and Settings\Mr. Evil\Desktop\lalsetup250.exe; C:\Documents and Settings\Mr. Evil\Desktop\netstumblerinstaller_0_4_0.exe. This is a deletion and staging lead only; corroborate with filesystem, registry, event-log, or network evidence before asserting broader activity.

- `tool_call_id`: `tc-135`
- artifact: `/home/assessor/.findevil/cases/bb2953cd-bdd9-4251-a493-70c97663f892/extracted/disk/disk-extract-affa2c27-51ae-4505-912a-27a7ec2a93a8/recyclebin/RECYCLER/S-1-5-21-2000478354-688789844-1708537768-1003/INFO2`
- confidence: Hypothesis — a single-source triage lead; corroborate before any response action.

FINDING 26 — CONFIDENCE: HYPOTHESIS, POOL: B, MITRE: T1074.001, REPLAY: EXACT_MATCH (MATCH)

hypothesis: NTUSER.DAT shellbag entries record user folder navigation to staging/tooling locations: \\4.12.220.254\Temp (registry_query, last_write 2004-08-26T15:06:50Z). Shellbags persist that a user browsed these folders in Explorer — here including a network staging share and tool directories. Navigation shows interest/access only; corroborate with file-system and timeline artifacts.

- `tool_call_id`: `tc-178`
- artifact: `/home/assessor/.findevil/cases/bb2953cd-bdd9-4251-a493-70c97663f892/extracted/disk/disk-extract-affa2c27-51ae-4505-912a-27a7ec2a93a8/registry/Documents and Settings/Mr. Evil/NTUSER.DAT`
- confidence: Hypothesis — a single-source triage lead; corroborate before any response action.

FINDING 27 — CONFIDENCE: HYPOTHESIS, POOL: B, MITRE: T1074.001, REPLAY: EXACT_MATCH (MATCH)

hypothesis: NTUSER.DAT shellbag entries record user folder navigation to staging/tooling locations: mIRC, Whois, channels, ftp, mirror.sg.depaul.edu, temp, ethereal, TEMPON~1.254, \\4.12.220.254\Temp (registry_query, last_write 2004-08-20T15:10:06Z). Shellbags persist that a user browsed these folders in Explorer — here including a network staging share and tool directories. Navigation shows interest/access only; corroborate with file-system and timeline artifacts.

- `tool_call_id`: `tc-179`
- artifact: `/home/assessor/.findevil/cases/bb2953cd-bdd9-4251-a493-70c97663f892/extracted/disk/disk-extract-affa2c27-51ae-4505-912a-27a7ec2a93a8/registry/Documents and Settings/Mr. Evil/NTUSER.DAT`
- confidence: Hypothesis — a single-source triage lead; corroborate before any response action.

FULL EVENT TIMELINE

Normalized timeline events: 627. First 40 rows shown below (consecutive identical events collapsed with an [Nx] count); full data is in `timeline.json` and analyst-friendly `timeline.csv`.

UTC TIME	ARTIFACT	EVENT	ACCOUNT	HOST	SOURCE IP	LOGON	PROCESS/PID	CONF.	TOOL CALL
1601-01-01T00:00:00Z	mft	mft entry: $Secure	—	—	—	—	—	HYPOTHESIS	tc-004
1999-04-23T22:22:00Z	mft	mft entry: WIN98/BASE4.CAB	—	—	—	—	—	HYPOTHESIS	tc-004
1999-04-23T22:22:00Z	mft	mft entry: WIN98/BASE5.CAB	—	—	—	—	—	HYPOTHESIS	tc-004
1999-04-23T22:22:00Z	mft	mft entry: WIN98/BASE6.CAB	—	—	—	—	—	HYPOTHESIS	tc-004
1999-04-23T22:22:00Z	mft	mft entry: WIN98/CATALOG3.CAB	—	—	—	—	—	HYPOTHESIS	tc-004
1999-04-23T22:22:00Z	mft	mft entry: WIN98/CHL99.CAB	—	—	—	—	—	HYPOTHESIS	tc-004
1999-04-23T22:22:00Z	mft	mft entry: WIN98/DELTEMP.COM	—	—	—	—	—	HYPOTHESIS	tc-004
1999-04-23T22:22:00Z	mft	mft entry: WIN98/DOSSETUP.BIN	—	—	—	—	—	HYPOTHESIS	tc-004
1999-04-23T22:22:00Z	mft	mft entry: WIN98/DRIVER1	—	—	—	—	—	HYPOTHESIS	tc-004

UTC TIME	ARTIFACT	EVENT	ACCOUNT	HOST	SOURCE IP	LOGON	PROCESS/PID	CONF.	TOOL CALL
		1.CAB							
1999-04-23T22:22:00Z	mft	mft entry: WIN98/DRIVER12.CAB	—	—	—	—	—	HYPOTHESIS	tc-004
1999-04-23T22:22:00Z	mft	mft entry: WIN98/DRIVER13.CAB	—	—	—	—	—	HYPOTHESIS	tc-004
1999-04-23T22:22:00Z	mft	mft entry: WIN98/DRIVER14.CAB	—	—	—	—	—	HYPOTHESIS	tc-004
1999-04-23T22:22:00Z	mft	mft entry: WIN98/DRIVER15.CAB	—	—	—	—	—	HYPOTHESIS	tc-004
1999-04-23T22:22:00Z	mft	mft entry: WIN98/DRIVER16.CAB	—	—	—	—	—	HYPOTHESIS	tc-004
1999-04-23T22:22:00Z	mft	mft entry: WIN98/DRIVER17.CAB	—	—	—	—	—	HYPOTHESIS	tc-004
1999-04-23T22:22:00Z	mft	mft entry: WIN98/DRIVER18.CAB	—	—	—	—	—	HYPOTHESIS	tc-004
1999-04-23T22:22:00Z	mft	mft entry: WIN98/DRIVER19.CAB	—	—	—	—	—	HYPOTHESIS	tc-004
1999-04-23T22:22:00Z	mft	mft entry: WIN98/DRIVER20.CAB	—	—	—	—	—	HYPOTHESIS	tc-004
1999-04-23T22:22:00Z	mft	mft entry: WIN98/EXTRACT.EXE	—	—	—	—	—	HYPOTHESIS	tc-004
1999-04-23T22:22:00Z	mft	mft entry: WIN98/FORMAT.COM	—	—	—	—	—	HYPOTHESIS	tc-004
1999-04-23T22:22:00Z	mft	mft entry: WIN98/INTL.TXT	—	—	—	—	—	HYPOTHESIS	tc-004
1999-04-23T22:22:00Z	mft	mft entry: WIN98/MINI.CAB	—	—	—	—	—	HYPOTHESIS	tc-004
1999-04-23T22:22:00Z	mft	mft entry: WIN98/NET10.CAB	—	—	—	—	—	HYPOTHESIS	tc-004
1999-04-23T22:22:00Z	mft	mft entry: WIN98/NET7.CAB	—	—	—	—	—	HYPOTHESIS	tc-004
1999-04-23T22:22:00Z	mft	mft entry: WIN98/NET8.CAB	—	—	—	—	—	HYPOTHESIS	tc-004
1999-04-23T22:22:00Z	mft	mft entry: WIN98/NET9.CAB	—	—	—	—	—	HYPOTHESIS	tc-004
1999-04-23T22:22:00Z	mft	mft entry: WIN98/OEMSETUP.BIN	—	—	—	—	—	HYPOTHESIS	tc-004
1999-04-23T22:22:00Z	mft	mft entry: WIN98/OEMSETUP.EXE	—	—	—	—	—	HYPOTHESIS	tc-004
1999-04-23T22:22:00Z	mft	mft entry: WIN98/PRECOPY1.CAB	—	—	—	—	—	HYPOTHESIS	tc-004
1999-04-23T22:22:00Z	mft	mft entry: WIN98/PRECOPY2.CAB	—	—	—	—	—	HYPOTHESIS	tc-004
1999-04-23T22:22:00Z	mft	mft entry: WIN98/SAVE32.COM	—	—	—	—	—	HYPOTHESIS	tc-004

UTC TIME	ARTIFACT	EVENT	ACCOUNT	HOST	SOURCE IP	LOGON	PROCESS/PID	CONF.	TOOL CALL
1999-04-23T22:22:00Z	mft	mft entry: WIN98/SCANDISK.EXE	—	—	—	—	—	HYPOTHESIS	tc-004
1999-04-23T22:22:00Z	mft	mft entry: WIN98/SCANDISK.PIF	—	—	—	—	—	HYPOTHESIS	tc-004
1999-04-23T22:22:00Z	mft	mft entry: WIN98/SCANPROG.EXE	—	—	—	—	—	HYPOTHESIS	tc-004
1999-04-23T22:22:00Z	mft	mft entry: WIN98/SCANREG.EXE	—	—	—	—	—	HYPOTHESIS	tc-004
1999-04-23T22:22:00Z	mft	mft entry: WIN98/SETUP.EXE	—	—	—	—	—	HYPOTHESIS	tc-004
1999-04-23T22:22:00Z	mft	mft entry: WIN98/SETUP.TXT	—	—	—	—	—	HYPOTHESIS	tc-004
1999-04-23T22:22:00Z	mft	mft entry: WIN98/SETUP0.WAV	—	—	—	—	—	HYPOTHESIS	tc-004
1999-04-23T22:22:00Z	mft	mft entry: WIN98/SETUP1.WAV	—	—	—	—	—	HYPOTHESIS	tc-004
1999-04-23T22:22:00Z	mft	mft entry: WIN98/SETUP2.WAV	—	—	—	—	—	HYPOTHESIS	tc-004

INDICATORS OF COMPROMISE (IOCS)

Observed artifacts pulled from the timeline and findings; validate or corroborate before detection deployment or blocking.

TYPE	VALUES
IP addresses	4.12.220.254
Domains	mirror.sg.depaul.edu
File paths	C:\Documents and Settings\Mr. Evil\Desktop\WinPcap_3_01_a.exe' (Software\Microsoft\Windows\CurrentVersion\Explorer\ComDlg32\OpenSaveMRU\, C:\Documents and Settings\Mr. Evil\Desktop\WinPcap_3_01_a.exe; C, C:\Documents and Settings\Mr. Evil\Desktop\ethereal-setup-0.10.6.exe' (Software\Microsoft\Windows\CurrentVersion\Explorer\ComDlg32\OpenSaveMRU\, C:\Documents and Settings\Mr. Evil\Desktop\lalsetup250.exe' (Software\Microsoft\Windows\CurrentVersion\Explorer\ComDlg32\OpenSaveMRU\, C:\Documents and Settings\Mr. Evil\Desktop\lalsetup250.exe; C, C:\Documents and Settings\Mr. Evil\Desktop\netstumblerinstaller_0_4_0.exe' (Software\Microsoft\Windows\CurrentVersion\Explorer\ComDlg32\OpenSaveMRU\

MITRE ATT&CK COVERAGE

6/12 ATT&CK targets covered by typed-tool output; 1 target(s) produced finding-level evidence; 5 target(s) remain blind spots

TECHNIQUE	TACTIC	STATUS	TOOLS OBSERVED	GAP / ANALYST VALUE
T1014 Rootkit	Defense Evasion	blind spot	none	missing or untouched artifact classes: memory
T1055 Process Injection	Defense Evasion / Privilege Escalation	blind spot	none	missing or untouched artifact classes: memory
T1059.001 PowerShell	Execution	covered, no finding (limited)	prefetch_parse	limited coverage — not proof of absence
T1021.001 Remote Desktop Protocol	Lateral Movement	blind spot	none	missing or untouched artifact classes: evtx
T1078 Valid Accounts	Defense Evasion / Persistence / Privilege Escalation	covered, no finding (limited)	registry_query	limited coverage — not proof of absence
T1003 OS Credential Dumping	Credential Access	available, not examined	none	required evidence class was available but no target tool ran
T1105 Ingress Tool Transfer	Command and Control	covered, no finding (limited)	mft_timeline	limited coverage — not proof of absence

TECHNIQUE	TACTIC	STATUS	TOOLS OBSERVED	GAP / ANALYST VALUE
T1071.001 Web Protocols	Command and Control	finding (CONFIRMED)	none	finding-level evidence exists; preserve cited tool output
T1071.004 DNS	Command and Control	blind spot	none	missing or untouched artifact classes: network
T1041 Exfiltration Over C2 Channel	Exfiltration	blind spot	none	missing or untouched artifact classes: network
T1547.001 Registry Run Keys / Startup Folder	Persistence / Privilege Escalation	covered, no finding (limited)	mft_timeline, prefetch_parse, registry_query	limited coverage — not proof of absence
T1053.005 Scheduled Task	Execution / Persistence / Privilege Escalation	covered, no finding (limited)	registry_query	limited coverage — not proof of absence

EVIDENCE COVERAGE

1/5 artifact classes touched; 1/5 directly available from supplied evidence

ARTIFACT CLASS	AVAILABLE	TOUCHED	TOOLS	CONFIDENCE IMPACT
memory	no	no	none	not a memory image; no live-process evidence
evtx	no	no	none	no event log supplied in this single-evidence run
disk/filesystem	yes	yes	disk_extract_artifacts, disk_mount, mft_timeline, prefetch_parse, registry_query	disk image registered; deep filesystem parsing requires mounted artifacts
network	no	no	none	no PCAP, Zeek, firewall, DNS, or proxy logs supplied
velociraptor	no	no	none	no Velociraptor collection supplied

ANALYSIS COVERAGE BY DOMAIN

This table shows which DFIR analysis domains the typed tools exercised on the supplied evidence. Coverage is scope, not assurance.

DOMAIN	STATUS	ARTIFACTS SEEN	TOOLS RUN	DATA SOURCES	GAPS
Host & Endpoint Forensics	automated	disk/filesystem	disk_extract_artifacts, mft_timeline, prefetch_parse, registry_query	DS0009, DS0022, DS0024	none
Endpoint Telemetry & Live Response	not_covered	none	none	none	missing or untouched artifact classes: velociraptor
Malware Analysis & Triage	partial	disk/filesystem	none	none	missing or untouched artifact classes: memory
Memory Forensics	not_covered	none	none	none	missing or untouched artifact classes: memory
Network Forensics	not_covered	none	none	none	missing or untouched artifact classes: network, no PCAP, Zeek, proxy, DNS, firewall, or NetFlow telemetry supplied
Windows Event & Account Analysis	not_covered	none	none	none	missing or untouched artifact classes: evtx

Overclaim guardrails applied: covered_no_finding is limited coverage, not a clean/cleared claim, Domain coverage describes triage/orchestration across the typed tools that ran, not certified-analyst judgment, visual exhibits do not create findings or upgrade confidence, execution claims still require at least two artifact classes

FIGURES

Visual exhibits are generated from parsed tool outputs. They support cited findings but do not replace `tool_call_id`-backed evidence or upgrade confidence by themselves.

HYPOTHESIS

- Card: `evidence-card-001`
- Linked findings: `f-A-mft-tools`
- Tool call: `tc-004`
- Source records: `mft_timeline:1, mft_timeline:2, mft_timeline:3`
- Confidence: `HYPOTHESIS`
- Citations: `CITE-MITRE-ATTACK-DATASOURCES`
- Why suspicious/relevant: This observable is relevant because finding ‘f-A-mft-tools’ is backed by parsed tool output ‘tc-004’ and should be interpreted with the cited artifact and source caveats.
- Snippet: `hypothesis: Hacking-tool artifacts present on disk as downloaded applications, recovered from the MFT: Program Files/Anonymizer \(created 2004-08-20T15:05:06Z\). The filesystem records \(MFT\) show these tool files in Program Files / on the use`
- Caveats: Visual exhibit supports the cited finding but does not replace parsed tool output., **HYPOTHESIS** confidence requires additional artifact corroboration.

HYPOTHESIS

- Card: `evidence-card-002`
- Linked findings: `f-A-mru-lalsetup250-exe`
- Tool call: `tc-176`
- Source records: `registry_query:562, registry_query:608, registry_query:609`
- Confidence: `HYPOTHESIS`
- Citations: `CITE-MITRE-ATTACK-DATASOURCES`
- Why suspicious/relevant: This observable is relevant because finding ‘f-A-mru-lalsetup250-exe’ is backed by parsed tool output ‘tc-176’ and should be interpreted with the cited artifact and source caveats.
- Snippet: `hypothesis: User recently opened-file MRU records 'C:\\Documents and Settings\\Mr. Evil\\Desktop\\lalsetup250.exe' \(Software\\Microsoft\\Windows\\CurrentVersion\\Explorer\\ComDlg32\\OpenSaveMRU\\*, registry_query, last_write 2004-08-27T15:18:05Z\). INF`
- Caveats: Visual exhibit supports the cited finding but does not replace parsed tool output., **HYPOTHESIS** confidence requires additional artifact corroboration.

HYPOTHESIS

- Card: `evidence-card-003`
- Linked findings: `f-A-mru-netstumblerinstaller-0-4-0-exe`
- Tool call: `tc-176`
- Source records: `registry_query:562, registry_query:608, registry_query:609`
- Confidence: `HYPOTHESIS`
- Citations: `CITE-MITRE-ATTACK-DATASOURCES`
- Why suspicious/relevant: This observable is relevant because finding ‘f-A-mru-netstumblerinstaller-0-4-0-exe’ is backed by parsed tool output ‘tc-176’ and should be interpreted with the cited artifact and source caveats.
- Snippet: `hypothesis: User recently opened-file MRU records 'C:\\Documents and Settings\\Mr. Evil\\Desktop\\netstumblerinstaller_0_4_0.exe' \(Software\\Microsoft\\Windows\\CurrentVersion\\Explorer\\ComDlg32\\OpenSaveMRU\\*, registry_query, last_write 2004-08-27T`
- Caveats: Visual exhibit supports the cited finding but does not replace parsed tool output., **HYPOTHESIS** confidence requires additional artifact corroboration.

HYPOTHESIS

- Card: `evidence-card-004`
- Linked findings: `f-A-mru-ethereal-setup-0-10-6-exe`
- Tool call: `tc-176`
- Source records: `registry_query:562, registry_query:608, registry_query:609`
- Confidence: `HYPOTHESIS`
- Citations: `CITE-MITRE-ATTACK-DATASOURCES`
- Why suspicious/relevant: This observable is relevant because finding ‘f-A-mru-ethereal-setup-0-10-6-exe’ is backed by parsed tool output ‘tc-176’ and should be interpreted with the cited artifact and source caveats.
- Snippet: `hypothesis: User recently opened-file MRU records 'C:\\Documents and Settings\\Mr. Evil\\Desktop\\ethereal-setup-0.10.6.exe' \(Software\\Microsoft\\Windows\\CurrentVersion\\Explorer\\ComDlg32\\OpenSaveMRU\\*, registry_query, last_write 2004-08-27T15:18`
- Caveats: Visual exhibit supports the cited finding but does not replace parsed tool output., **HYPOTHESIS** confidence requires additional artifact corroboration.

HYPOTHESIS

- Card: `evidence-card-005`
- Linked findings: `f-A-mru-winpcap-3-01-a-exe`
- Tool call: `tc-176`
- Source records: `registry_query:562, registry_query:608, registry_query:609`
- Confidence: `HYPOTHESIS`
- Citations: `CITE-MITRE-ATTACK-DATASOURCES`
- Why suspicious/relevant: This observable is relevant because finding ‘f-A-mru-winpcap-3-01-a-exe’ is backed by parsed tool output ‘tc-176’ and should be interpreted with the cited artifact and source caveats.

- Snippet: `hypothesis: User recently opened-file MRU records 'C:\\Documents and Settings\\Mr. Evil\\Desktop\\WinPcap_3_01_a.exe' \(Software\\Microsoft\\Windows\\CurrentVersion\\Explorer\\ComDlg32\\OpenSaveMRU\\*, registry_query, last_write 2004-08-27T15:18:05Z\).`
- Caveats: Visual exhibit supports the cited finding but does not replace parsed tool output., **HYPOTHESIS** confidence requires additional artifact corroboration.

HYPOTHESIS

- Card: `evidence-card-006`
- Linked findings: `f-A-sam-mr-evil`
- Tool call: `tc-206`
- Source records: `registry_query:360, registry_query:361, registry_query:367`
- Confidence: `HYPOTHESIS`
- Citations: `CITE-MITRE-ATTACK-DATASOURCES`
- Why suspicious/relevant: This observable is relevant because finding ‘f-A-sam-mr-evil’ is backed by parsed tool output ‘tc-206’ and should be interpreted with the cited artifact and source caveats.
- Snippet: `hypothesis: User account 'Mr. Evil' with suspicious naming was created on this system: it is recorded in the SAM \(Security Account Manager\) hive \(SAM\\Domains\\Account\\Users\\Names\\Mr. Evil; the Names subkey last_write 2004-08-19T23:03:54Z app`
- Caveats: Visual exhibit supports the cited finding but does not replace parsed tool output., **HYPOTHESIS** confidence requires additional artifact corroboration.

CAIN.EXE EXECUTED ON THIS HOST

- Card: `evidence-card-007`
- Linked findings: `f-B-prefetch-cain-exe`
- Tool call: `tc-010`
- Source records: `prefetch_parse:615`
- Confidence: `CONFIRMED`
- Citations: `CITE-MITRE-ATTACK-DATASOURCES`
- Why suspicious/relevant: This observable is relevant because finding ‘f-B-prefetch-cain-exe’ is backed by parsed tool output ‘tc-010’ and should be interpreted with the cited artifact and source caveats.
- Snippet: `cain.exe executed on this host: Windows Prefetch records its execution and the UserAssist key \(per-user GUI execution\) records the same binary. Two independent artifact classes \(prefetch + registry/UserAssist\) corroborate execution.`
- Caveats: Visual exhibit supports the cited finding but does not replace parsed tool output.

CAIN25B45.EXE EXECUTED ON THIS HOST

- Card: `evidence-card-008`
- Linked findings: `f-B-prefetch-cain25b45-exe`
- Tool call: `tc-011`
- Source records: `prefetch_parse:521`
- Confidence: `CONFIRMED`
- Citations: `CITE-MITRE-ATTACK-DATASOURCES`
- Why suspicious/relevant: This observable is relevant because finding ‘f-B-prefetch-cain25b45-exe’ is backed by parsed tool output ‘tc-011’ and should be interpreted with the cited artifact and source caveats.
- Snippet: `cain25b45.exe executed on this host: Windows Prefetch records its execution and the UserAssist key \(per-user GUI execution\) records the same binary. Two independent artifact classes \(prefetch + registry/UserAssist\) corroborate execution.`
- Caveats: Visual exhibit supports the cited finding but does not replace parsed tool output.

ETHEREAL-SETUP-0.10.6.EXE EXECUTED ON THIS HOST

- Card: `evidence-card-009`
- Linked findings: `f-B-prefetch-ethereal-setup-0-10-6-exe`
- Tool call: `tc-020`
- Source records: `prefetch_parse:612`
- Confidence: `CONFIRMED`
- Citations: `CITE-MITRE-ATTACK-DATASOURCES`
- Why suspicious/relevant: This observable is relevant because finding ‘f-B-prefetch-ethereal-setup-0-10-6-exe’ is backed by parsed tool output ‘tc-020’ and should be interpreted with the cited artifact and source caveats.
- Snippet: `ethereal-setup-0.10.6.exe executed on this host: Windows Prefetch records its execution and the UserAssist key \(per-user GUI execution\) records the same binary. Two independent artifact classes \(prefetch + registry/UserAssist\) corroborate e`
- Caveats: Visual exhibit supports the cited finding but does not replace parsed tool output.

ETHEREAL.EXE EXECUTED ON THIS HOST

- Card: `evidence-card-010`
- Linked findings: `f-B-prefetch-ethereal-exe`
- Tool call: `tc-021`
- Source records: `prefetch_parse:616`
- Confidence: `CONFIRMED`
- Citations: `CITE-MITRE-ATTACK-DATASOURCES`
- Why suspicious/relevant: This observable is relevant because finding ‘f-B-prefetch-ethereal-exe’ is backed by parsed tool output ‘tc-021’ and should be interpreted with the cited artifact and source caveats.
- Snippet: `ethereal.exe executed on this host: Windows Prefetch records its execution and the UserAssist key \(per-user GUI execution\) records the same binary. Two independent artifact classes \(prefetch + registry/UserAssist\) corroborate execution.`

- Caveats: Visual exhibit supports the cited finding but does not replace parsed tool output.

REFERENCES

CITATION ID	TITLE	URL	SUPPORTS
CITE-MITRE-ATTACK-DATASOURCES	MITRE ATT&CK Data Sources	https://attack.mitre.org/datasources/	ATT&CK data-source coverage mapping
CITE-MITRE-T1003-001	MITRE ATT&CK T1003.001 LSASS Memory	https://attack.mitre.org/techniques/T1003/001/	LSASS credential-dumping interpretation
CITE-MITRE-T1014	MITRE ATT&CK T1014 Rootkit	https://attack.mitre.org/techniques/T1014/	DKOM/rootkit process-view divergence interpretation
CITE-NIST-800-61R2	NIST SP 800-61 Rev. 2 Computer Security Incident Handling Guide	https://csrc.nist.gov/pubs/sp/800/61/r2/final	separation of evidence, analysis, response actions, and gaps
CITE-PLASO	Plaso/log2timeline documentation	https://plaso.readthedocs.io/	multi-source forensic timeline normalization
CITE-TIMESKETCH	Timesketch documentation	https://timesketch.org/	analyst-oriented forensic timeline review
CITE-VOLATILITY3	Volatility 3 documentation	https://volatility3.readthedocs.io/	memory plugin output and process-view validation
CITE-ZEEK-LOGS	Zeek log documentation	https://docs.zeek.org/en/current/logs/index.html	network log and protocol-semantic coverage
CITE-VELOCIRAPTOR-ARTIFACTS	Velociraptor artifact documentation	https://docs.velociraptor.app/docs/artifacts/	artifact-based endpoint collection
CITE-SIGMAHQ	SigmaHQ rules repository	https://github.com/SigmaHQ/sigma	structured log detection rules as triage leads
CITE-HAYABUSA	Hayabusa repository	https://github.com/Yamato-Security/hayabusa	Windows EVTX timeline and hunting output
CITE-CAPA	capa repository	https://github.com/mandiant/capa	malware capability triage limits

RECOMMENDED ANALYST ACTIONS

PRIORITY	ACTION	WHY	BASED ON	EXPECTED EVIDENCE
P2	Collect Security, Sysmon, and PowerShell Operational EVTX and rerun EVTX/Hayabusa analysis.	Current findings lack event-log corroboration for logon, process creation, and PowerShell execution hypotheses.	evtx_gap	Security 4624/4625/4688, Sysmon 1/3/7/10/11, PowerShell 4103/4104
P2	Use the disk artifact summary to pivot between Prefetch, Registry, MFT, USN, EVTX, and YARA-target rows without upgrading single-source execution claims.	Extracted disk artifacts are now summarized as leads and timeline context; execution wording still needs two artifact classes and cited tool_call_id evidence.	disk_artifact_summary	Correlated Prefetch run times, Registry LastWrite, MFT/USN timestamps, EVTX records, and YARA hits
P3	Acquire DNS, proxy, firewall, NetFlow, or PCAP telemetry to test C2 and exfiltration hypotheses.	Network telemetry was not supplied or parsed in this run, so exfiltration and command-and-control coverage remains a blind spot.	network_gap	DNS queries, proxy URLs, firewall sessions, PCAP, Velociraptor network collection
P3	Close ATT&CK blind spots before making closure decisions.	The coverage matrix identifies target techniques with no supporting artifact class in this run.	T1014, T1055, T1021.001, T1071.004, T1041	Additional evidence classes mapped in attack_coverage.targets[].artifact_classes
P4	Pivot from the first and last normalized timeline events into adjacent artifact classes.	Temporal clustering often exposes execution chains that a single artifact class cannot prove alone.	timeline	timeline.csv plus adjacent EVTX, Prefetch, MFT, and network events

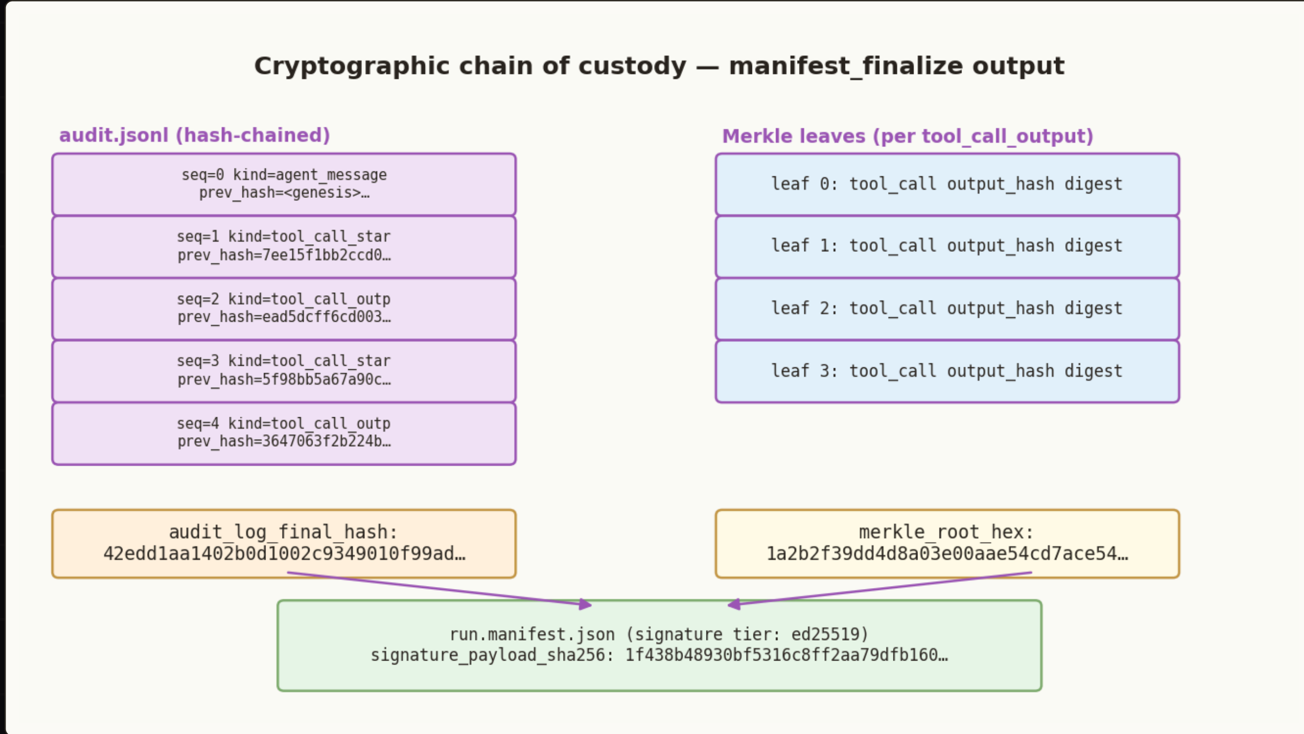
REPRODUCIBILITY APPENDIX

Verifier replay artifacts record whether each cited tool call reproduced the audited output hash. They do not change Track 3b severity policy.

FINDING	TOOL	DRIFT CLASS	MATCH	EXPECTED SHA	ACTUAL SHA
f-A-mft-tools	mft_timeline	exact_match	yes	9dbbbd01e495	9dbbbd01e495
f-A-mru-lalsetup250-exe	registry_query	exact_match	yes	2c1a21bddeea	2c1a21bddeea

FINDING	TOOL	DRIFT CLASS	MATCH	EXPECTED SHA	ACTUAL SHA
f-A-mru-netstumblerinstaller-0-4-0-exe	registry_query	exact_match	yes	2c1a21bddeea	2c1a21bddeea
f-A-mru-ethereal-setup-0-10-6-exe	registry_query	exact_match	yes	2c1a21bddeea	2c1a21bddeea
f-A-mru-winpcap-3-01-a-exe	registry_query	exact_match	yes	2c1a21bddeea	2c1a21bddeea
f-A-sam-mr-evil	registry_query	exact_match	yes	0bdce70657b4	0bdce70657b4
f-B-prefetch-cain-exe	prefetch_parse	exact_match	yes	b2e14c91aaa2	b2e14c91aaa2
f-B-prefetch-cain25b45-exe	prefetch_parse	exact_match	yes	76ae3ea8345e	76ae3ea8345e
f-B-prefetch-ethereal-setup-0-10-6-exe	prefetch_parse	exact_match	yes	eba2574fcda2	eba2574fcda2
f-B-prefetch-ethereal-exe	prefetch_parse	exact_match	yes	cf2fcec98191	cf2fcec98191
f-B-prefetch-lookatlan-exe	prefetch_parse	exact_match	yes	9a10d59745b3	9a10d59745b3
f-B-prefetch-mirc-exe	prefetch_parse	exact_match	yes	b035f2359a05	b035f2359a05
f-B-prefetch-mirc612-exe	prefetch_parse	exact_match	yes	b36f1b611dd3	b36f1b611dd3
f-B-prefetch-netstumbler-exe	prefetch_parse	exact_match	yes	c912d93baf9b	c912d93baf9b
f-B-prefetch-netstumblerinstaller-0-4-0-ex	prefetch_parse	exact_match	yes	80850ecc397c	80850ecc397c
f-B-lnk-removable-media-7e41075f	ez_parse	exact_match	yes	cd5f7a5e5216	cd5f7a5e5216
f-B-lnk-removable-media-424a9a13	ez_parse	exact_match	yes	ae577890c3bc	ae577890c3bc
f-B-lnk-removable-media-9eddb58a	ez_parse	exact_match	yes	5e8cd9339136	5e8cd9339136
f-B-lnk-removable-media-ee34c604	ez_parse	exact_match	yes	21330f99dddb	21330f99dddb
f-B-lnk-removable-media-dd676bb7	ez_parse	exact_match	yes	d33d24d6da95	d33d24d6da95
f-B-lnk-removable-media-191ee7e5	ez_parse	exact_match	yes	c94d3626c65f	c94d3626c65f
f-B-lnk-removable-media-9710644c	ez_parse	exact_match	yes	eda83e9a9f9b	eda83e9a9f9b
f-B-lnk-removable-media-00082918	ez_parse	exact_match	yes	410b85dabb71	410b85dabb71
f-B-lnk-removable-media-c9f8b23e	ez_parse	exact_match	yes	72c228fde4e5	72c228fde4e5
f-B-recyclebin-staging	plaso_parse	exact_match	yes	4fa78692829f	4fa78692829f
f-B-shellbag-06b9e73e	registry_query	exact_match	yes	acfb334a3954	acfb334a3954
f-B-shellbag-3c13f240	registry_query	exact_match	yes	456cb1bf6660	456cb1bf6660

CHAIN OF CUSTODY



Chain of custody

INTEGRITY VERIFICATION

This investigation produced a `run.manifest.json` that any third party can verify offline from the VERDICT repository using the manifest verification library or the `manifest_verify` MCP tool. There is no standalone `manifest_verify` shell command in this repo.

```
uv run --directory services/agent python -c "from pathlib import Path; from findevil_agent.crypto.manifest import verify_manifest;
print(verify_manifest(Path('PATH/TO/run.manifest.json'), audit_log_path=Path('PATH/TO/audit.jsonl')))"
# returns overall=True if the audit chain, Merkle root, leaf count, and signature presence validate
```

The verifier rebuilds: 1. The audit chain by walking `prev_hash` SHA-256 links (catches backdated edits). 2. The Merkle tree from the manifest's `leaves[]` array (catches selective redaction). 3. The signature bundle recorded in the manifest. Ed25519 signatures verify offline in `manifest_verify`; Sigstore bundles are recorded for identity-policy-aware verification by a party that supplies the expected signer identity.

A tamper test against this manifest's `merkle_root_hex` was not run automatically. To execute it, copy the manifest, overwrite `merkle_root_hex` with `ff` repeated 32 times, then run the same Python verification command against the tampered copy.

```
python -c "import shutil;shutil.copyfile('run.manifest.json','run.manifest.tamper.json')"
python -c "import
json,pathlib;p=pathlib.Path('run.manifest.tamper.json');d=json.loads(p.read_text());d['merkle_root_hex']='ff'*32;p.write_text(json.dumps(d,indent=2,sort_keys=T
uv run --directory services/agent python -c "from pathlib import Path; from findevil_agent.crypto.manifest import verify_manifest;
print(verify_manifest(Path('PATH/TO/run.manifest.tamper.json'), audit_log_path=Path('PATH/TO/audit.jsonl')))"
```

Produced by `find-evil-auto` (the VERDICT automated investigation orchestrator). The cryptographic attestation values shown are the actual outputs of this run; every quantitative claim above is independently verifiable from the artifacts in this directory (`audit.jsonl`, `run.manifest.json`, `verdict.json`). The automated QA / expert-signoff gates for this run are in the companion `REPORT-internal` packet.